THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL UNIT TEST SET 2

Chapters 11–14: **Exploring Some Geometric Themes, Tales by Dots and Lines, Algebra Play, Area** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The area of a triangle is 120 cm². If the altitude on the side of 20 cm is 12 cm, the altitude on the side of 16 cm (in cm) is: 1

(a) 15 **(b)** 12 **(c)** 17 **(d)** 7

2. A Koch curve starts with a 81 cm line. The length of the curve at stage 4 (in cm) is: 1

(a) 256 **(b)** 324 **(c)** 337 **(d)** 229

3. A data set 54, 34, 23, 16, 31, 46, 57 gets an extra (outlier) value 540. The median of the new data of 8 values is: 1

(a) 100.12 **(b)** 540 **(c)** 34 **(d)** 40

4. The value of $18^2 - 15 \times 21$ is: 1

(a) 6 **(b)** 9 **(c)** 0 **(d)** 18

5. A rhombus has perimeter 116 cm and one diagonal 40 cm. Its area is: 1

(a) 1680 cm² **(b)** 840 cm² **(c)** 420 cm² **(d)** 1160 cm²

6. A Sierpinski triangle starts with a triangle of side 64 cm. At stage 4 the side of each small shaded triangle is (in cm): 1

(a) 0.79 **(b)** 4 **(c)** 8 **(d)** 5

7. A car travels 120 km at 40 km/h and then 120 km at 60 km/h. Its average speed (km/h) for the whole journey is: 1

(a) 24 **(b)** 90 **(c)** 48 **(d)** 50

8. The sum of three consecutive integers is 63. The largest of them is: 1

(a) 21 **(b)** 22 **(c)** 20 **(d)** 23

9. A 52 m by 36 m rectangular lawn has a path 4 m wide inside along its boundary. The area of the lawn left is: 1

(a) 1808 m² **(b)** 1536 m² **(c)** 1168 m² **(d)** 1232 m²

10. A polyhedron has 8 triangular faces and 2 quadrilateral faces and no other faces. Its number of vertices is: 1

(a) 10 **(b)** 28 **(c)** 8 **(d)** 16

11. The mean of 8 numbers is 15. If every number is multiplied by 3 and then 4 is added, the new mean is: 1

(a) 45 **(b)** 49 **(c)** 19 **(d)** 57

12. A three-digit number with digits a, b, c is reversed. If $a - c = 2$, the difference between the number and its reverse is: 1

(a) 180 **(b)** 200 **(c)** 198 **(d)** 18

13. A triangle has sides 33 cm, 56 cm and 65 cm. Its area is: 1

(a) 154 cm² **(b)** 924 cm² **(c)** 1072 cm² **(d)** 1848 cm²

14. All the 16 faces of a polyhedron are pentagons. Its number of vertices is: 1

(a) 24 **(b)** 58 **(c)** 40 **(d)** 26

15. The mean of 5 numbers is 24. When one number is removed the mean of the remaining 4 becomes 21. The removed number is: 1

(a) 36 **(b)** 3 **(c)** 24 **(d)** 9

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. The mean of the first 8 even natural numbers is 9. Verify this for n = 8. 2

17. If $x + y = 17$ and $x - y = 5$, find $x^2 - y^2$. 2

18. A rectangle's length is increased by 30% and breadth decreased by 20%. By what percent does the area change? (original area 100 units) 2

19. A Sierpinski triangle starts with a triangle of side 128 cm. Find the number of shaded triangles and the side of each at stage 5. 2

20. In a data set 39, 27, 21, 42, 20, 43 the value 39 was wrongly copied as 45. Find the correct mean. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. An isosceles trapezium has parallel sides 14 cm and 32 cm and each slant side 15 cm. Find its height and area. 3

22. In the Sierpinski triangle construction, find the first stage at which the number of shaded triangles exceeds 500. How many shaded triangles are there at that stage? 3

23. A student's marks in 5 tests are 49, 75, 64, 74, 48. What mark must she score in the 6th test so that her mean becomes 68? 3

24. The sum of the dates of a 2×2 block in a calendar is 36. Find the four dates and show that the block's sum is always $4\times$(smallest date)$+16$. 3

OR

A garden 40 m by 30 m has a rectangular pond 10 m by 8 m inside it and a path 2 m wide around the garden (outside). Find the area of the garden excluding the pond, and the area of the path. 3

25. Every face of a polyhedron is a triangle and it has 40 faces. Find the number of edges and vertices and verify Euler's formula. 3

OR

A class of 25 students has mean mark 48. After rechecking, a score of 20 was corrected to 70. Find the new mean. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. A Koch snowflake is built from an equilateral triangle of side 27 cm. Find its perimeter at stage 3, and by how much the perimeter has increased from stage 0. 5

OR

The mean of 12 observations is 15. If each observation is multiplied by 4 and then decreased by 5, find the new mean and the new sum. 5

27. A three-digit number has digits a, b, c with $a - c = 1$. Show that the number minus its reverse is divisible by 99 and find the difference. 5

OR

A trapezium ABCD has parallel sides AB = 14 cm and DC = 26 cm and height 10 cm. A parallelogram of base 14 cm and the same height is cut off from it, leaving a triangle. Find the area of the trapezium, the parallelogram and the remaining triangle. 5

END OF QUESTION PAPER Total: 50 Marks

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